18 Lecture 18

In Lecture 14, we studied some ways that the a sequence of random variables $(X_n)_{n\in\mathbb{N}}$ defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ can converge to a random variable X also defined on $(\Omega, \mathcal{F}, \mathbb{P})$. In this lecture, we will discuss a different type of convergence: *convergence in distribution.*

18.1 Convergence in distribution

Roughly speaking, convergence in distribution means that the sequence of probability measures $(\mathbb{P}^{X_n})_{n\in\mathbb{N}}$ induced by $(X_n)_{n\in\mathbb{N}}$ converges to the probability measure $\mathbb{P}^X$ in the appropriate sense.

Definition 18.1 (Convergence in distribution)**.** Let $(X_n)_{n\in\mathbb{N}}, X$ be random variables with values in $\mathbb{R}$. We say that $(X_n)_{n\in\mathbb{N}}$ converges in distribution to X and we write $X_n \xrightarrow{d} X$ if and only if

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x),$$

for any $x \in \mathbb{R}$ such that $F_X(x) = F_X(x-)$, that is, for any $x \in \mathbb{R}$ such that F_X is continuous at x.

Remark 18.2. If $(X_n)_{n\in\mathbb{N}}$ and X are defined on the same probability space, then it makes sense to study how convergence in distribution is related to the other types of convergence.

Example 1 (Exponential distribution)
Suppose that $X_n \sim \text{Geom}(1/n)$ for $n \in \mathbb{N}$. Let's show that $\frac{1}{n}X_n \xrightarrow{d} X \sim \text{Exp}(1)$.

We have to verify the equality

$$\lim_{n\to\infty} F_{X_n/n}(x) = F(x) = \begin{cases} 0, & \text{if } x \le 0, \\ 1 - e^{-x}, & \text{if } x > 0. \end{cases}$$

For $x \le 0$, we have $F_{X_n/n}(x) = \mathbb{P}(X_n/n \le x) = 0$ and $F_X(x) = 0$, so $\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$.
For $x > 0$, we compute

$$1 - F_{X_n/n}(x) = \mathbb{P}(X_n > nx) = \mathbb{P}(X_n > \lfloor nx \rfloor) = \left(1 - \frac{1}{n}\right)^{\lfloor nx \rfloor} = \left(\left(1 - \frac{1}{n}\right)^n\right)^{\lfloor nx \rfloor / n} \xrightarrow{n\to\infty} (e^{-1})^x = e^{-x},$$

because $\frac{nx-1}{n} \le \frac{\lfloor nx \rfloor}{n} \le \frac{nx}{n} = x$ and $\frac{nx-1}{n} \xrightarrow{n\to\infty} x$. We deduce that

$$\lim_{n\to\infty} F_{X_n/n}(x) = 1 - e^{-x} = F_X(x), \quad x > 0.$$

Interpretation: We flip a coin with probability $1/n$ for success, every $1/n$ seconds. We record X_n the first successful try and $\frac{1}{n}X_n$ the time of first success. Then, as $n \to \infty$ the distribution of the first successful try tends to be exponential.

Suggested problem: How would you modify the above observation so that the limit $X \sim \text{Exp}(\lambda)$, for some $\lambda > 0$?

Example 2
Let $(X_n)_{n\in\mathbb{N}}$ be sequence of i.i.d random variables such that $X_1 \sim \text{Exp}(1)$. For any $n \in \mathbb{N}$ we consider $W_n := \max\{X_1, \dots, X_n\}$ and $Z_n := W_n - \log n$.

Question: Study whether the sequence $(Z_n)_{n\in\mathbb{N}}$ converges in distribution. If it does, find the limiting distribution.

We first compute the distribution of Z_n. For $x \in \mathbb{R}$ we have

$$F_{Z_n}(x) = \mathbb{P}(W_n - \log n \le x) = \mathbb{P}(\max\{X_1 \dots, X_n\} \le x + \log n) = \mathbb{P}(X_1 \le x + \log n, \dots, X_n \le x + \log n)$$

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$$= (\mathbb{P}(X_1 \le x + \log n))^n = (1 - e^{-x-\log n})^n = \left(1 - \frac{e^{-x}}{n}\right)^n.$$

From this formula, we see that $\lim_{n\to\infty} F_{Z_n}(x) = e^{-e^{-x}}$. We observe that the function $F(x) = e^{-e^{-x}}$ is increasing, continuous

$$\lim_{x\to+\infty} F(x) = 1 \quad \text{and} \quad \lim_{x\to-\infty} F(x) = 0,$$

therefore it satisfies all the properties of a distribution function. This distribution is called Gumbel.

Proposition 18.3. *Let $(X_n)_{n\in\mathbb{N}}, X$ be random variables. Then, $X_n \xrightarrow{d} X$ if and only if*

$$\lim_{n\to\infty} \mathbb{E}[h(X_n)] = \mathbb{E}[h(X)],$$

for any $h\colon \mathbb{R} \to \mathbb{R}$ continuous and bounded function.

Example 3 (uniform distribution)
Assume that $(X_n)_{n\in\mathbb{N}}$ is a sequence of random variables such that X_n is uniform with values in the set $\{1, \dots, n\}$, that is $\mathbb{P}(X_n = k) = 1/n$ for all $k \in \{1, \dots, n\}$. We will show that

$$\frac{1}{n} X_n \xrightarrow{d} U \sim \text{Uniform}(0, 1).$$

For $h\colon \mathbb{R} \to \mathbb{R}$ bounded and continuous we have

$$\mathbb{E}\left[h\left(\frac{X_n}{n}\right)\right] = \sum_{k=1}^{n} \mathbb{P}(X_n = k) h\left(\frac{k}{n}\right) = \frac{1}{n}\sum_{k=1}^{n} h\left(\frac{k}{n}\right) \xrightarrow{n\to\infty} \int_0^1 h(x)dx = \mathbb{E}[h(U)],$$

because of the convergence of the Riemann sum to the Riemann integral.

Proposition 18.4. *Let $(X_n)_{n\in\mathbb{N}}, X$ be random variable with values in $\mathbb{R}$ such that $X_n \xrightarrow{d} X$. If $g\colon \mathbb{R} \to \mathbb{R}$ is continuous, then $g(X_n) \xrightarrow{d} g(X)$.*

Proof. Recall that, since g is continuous, $(g(X_n))_{n\in\mathbb{N}}$, $g(X)$ are random variables (Proposition 5.5 for $k = 1$). For any $h\colon \mathbb{R} \to \mathbb{R}$ bounded and continuous, $h \circ g$ is also bounded and continuous, therefore by Proposition 18.3 we have

$$\lim_{n\to\infty} \mathbb{E}[h(g(X_n))] = \mathbb{E}[h(g(X))].$$

Since this holds for any h, by Proposition 18.3, once again, we conclude $g(X_n) \xrightarrow{d} g(X)$. □

Theorem 18.5 (Convergence in probability implies convergence in distribution)**.** *Let $(X_n)_{n\in\mathbb{N}}, X$ be random variables defined on the same probability space. If $X_n \xrightarrow{\mathbb{P}} X$, then $X_n \xrightarrow{d} X$.*

Proof. We argue by contradiction. If $X_n \xrightarrow{d} X$ is not true, then there exists a subsequence $(X_{n_k})_{k\in\mathbb{N}}$ and a bounded and continuous function $h\colon \mathbb{R} \to \mathbb{R}$ such that

$$|\mathbb{E}[h(X_{n_k})] - \mathbb{E}[h(X)]| > \varepsilon, \text{ for some } \varepsilon > 0. \tag{18.1}$$

Since $X_{n_k} \xrightarrow{\mathbb{P}} X$, by Theorem 14.4, there exists a further subsequence $(X_{n_{k_r}})_{r\in\mathbb{N}}$ such that $X_{n_{k_r}} \xrightarrow{\text{a.s. } r\to\infty} X$. However, h is continuous, so $h(X_{n_{k_r}}) \xrightarrow{\text{a.s. } r\to\infty} h(X)$. Finally, h is bounded, hence the Dominated Convergence Theorem gives

$$\lim_{r\to\infty} \mathbb{E}[h(X_{n_{k_r}})] = \mathbb{E}[h(X)],$$

contradicting (18.1). □

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